Tree Health Monitoring Using Deep Learning: A Comparative Study of Models for Imbalanced Environmental Data

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GOAL

Automate the classification of tree health using deep learning, replacing the slow and time-consuming manual inspections. By using image data collected from a tree nursery, the system seeks to detect early signs of decline and stress in real time. Automating this process can enable faster intervention and provide valuable insights into tree health over time.

PROBLEM

Tree health monitoring is traditionally a manual task, that cannot keep up with the dynamic nature of plant conditions. By the time a tree is assessed, its health may have already declined further, making timely intervention difficult or ineffective:

- Manual inspection is slow and not scalable
- Delayed detection reduces opportunities for timely intervention
- No automated use of available visual data
- Lack of accessible AI tools for non-technical stakeholders

METHODOLOGY

To address the problem of delayed and inconsistent tree health monitoring, the following methods were used:

- Data analysis and cleaning
- Preprocessing (normalization, resizing, and data augmentation)
- Class imbalance handling strategies (oversampling and class weighting)
- Training
- Evaluation

$$Recall = \frac{True\ Positives + False\ Negatives}{True\ Positives}$$

$$F1\ Score = 2 \times \frac{Precision + Recall}{Precision \times Recall}$$

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives}$$

$$Accuracy = \frac{True\ Positives}{True\ Positives + False\ Positives + No\ Identifications}$$

DATA

The dataset used in this project originates from a real-world case study provided by Dr. Andrew Hirons at Myerscough College, UK.
It contains over **5,200 images** of **Acer and Liquidambar** trees. Each image is labeled with one of five phenological stages: **Dormancy, Leaf Flushing, Crown Development, Pruning Event, and Senescence**.

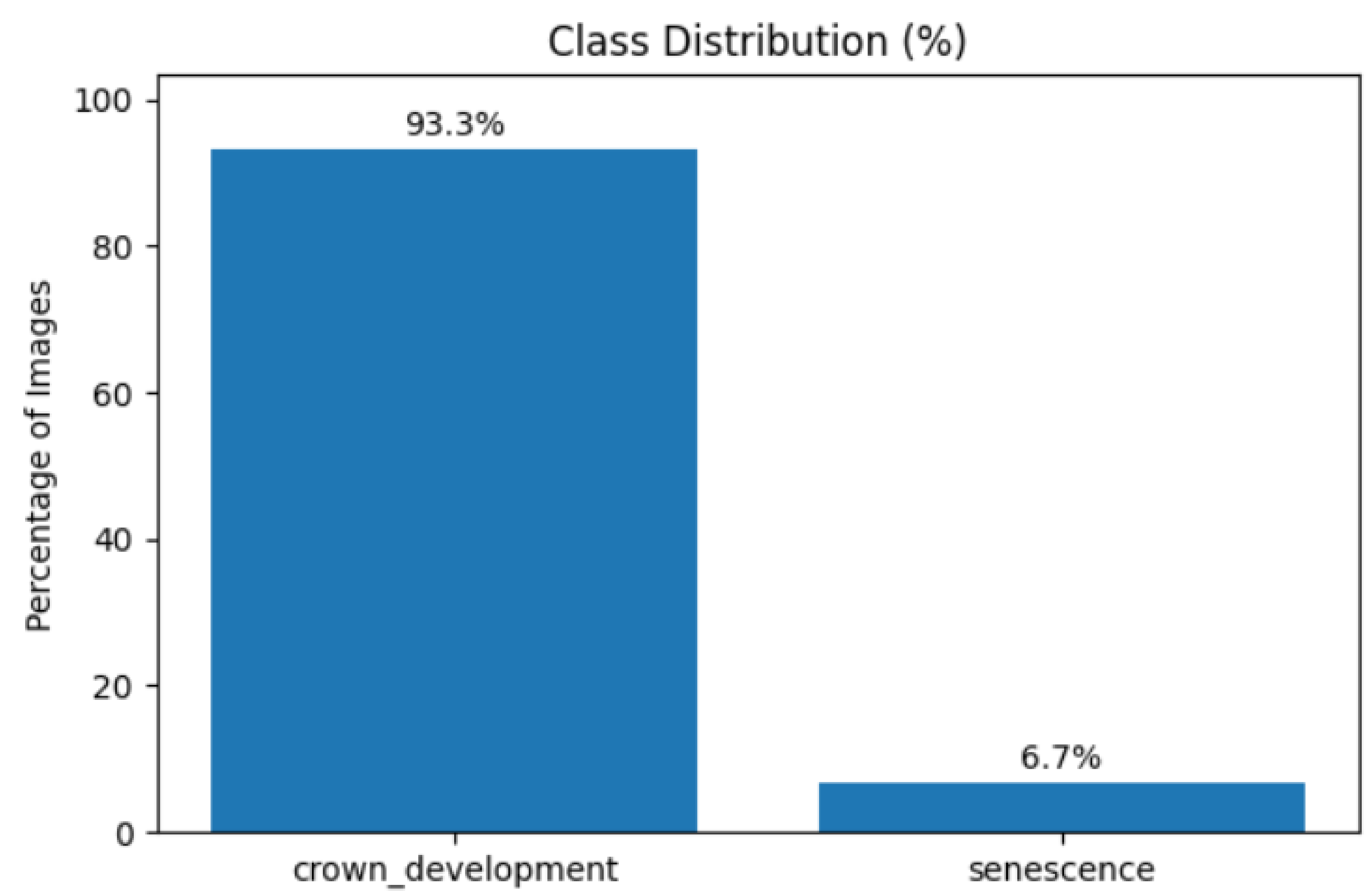
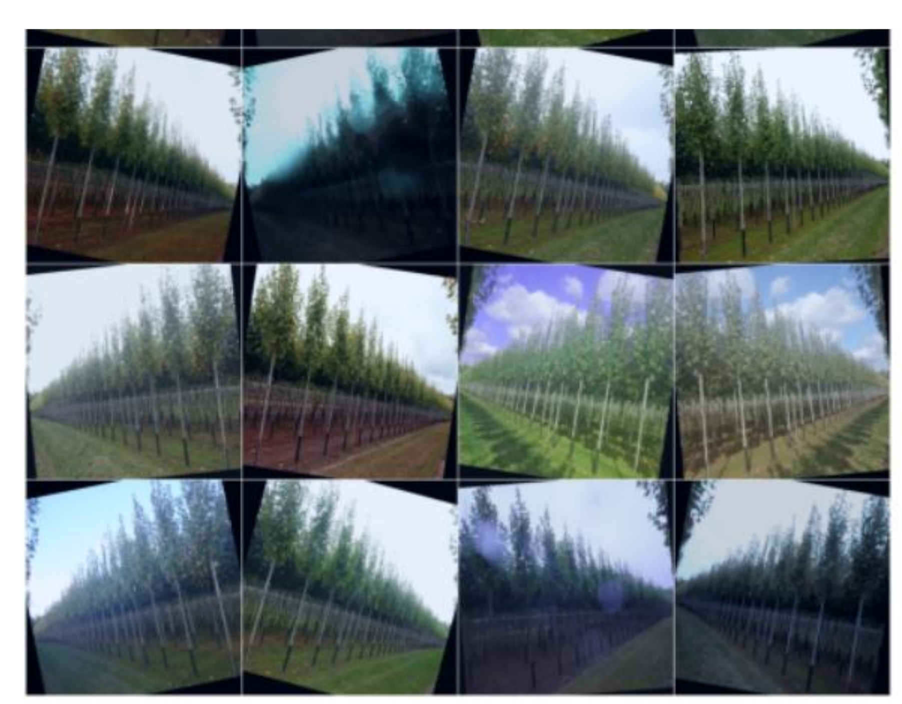


Figure | Class distribution

Some phenological stages represented less than 1% of the total samples. The task was scope down to: **Crown Development** (healthy) and **Senescence** (unhealthy). Only images of **Acer** trees were retained, as the Liquidambar samples contained too few instances and differed visually in structure. After filtering, a total of **1,667 images** remained.



```
Batch labels: ['Senescence', 'Senescence', 'Senescence', 'Senescence
'Senescence', 'Senescence', 'Senescence', 'Crown', 'Crown', 'Senescei
Batch distribution → Crown: 14, Senescence: 18
```

Figure | Batch distribution

MODELS

Following models were selected for this project due to their strong performance in imbalanced classification tasks.

- EfficientNetV2 - high accuracy with efficient computation
- Swin Transformer - strong global context modeling through attention mechanisms.

	EfficientNetV2 (CNN)	**Swin Transformer (ViT)**
Overall Accuracy	**High** (97-99%)	**High** (95-98%)
Handles Imbalanced Data	**Good** (with augmentation)	**Very Good** (attention mechanism)
Interpretability	**Good** (with heatmap)	**Excellent** (built-in attention)
Computational Efficiency	**Excellent** (lighter, scalable)	**Moderate** (heavier, resource-intensive)
Ease of Fine-tuning	**Excellent**	**Good**
Deployment & Scalability	**Excellent** (IoT, edge-friendly)	**Moderate** (less edge-friendly)
Community & Resources	**Excellent** (widely supported)	**Good** (growing community)

Table | Model's Comparison

NEXT STEPS

Models Training on the dataset using consistent experimental settings.

Compare the results based primarily on **recall**, to assess their ability to correctly detect unhealthy trees.

Design and implement a user-friendly dashboard to visualize tree health predictions, including class outputs.

SCAN ME

https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fraw.githubusercontent.com%2Fvalkov01%2FTreeVid%2Frefs%2Fheads%2Fmain%2FMidterm%2520Poster.pptx&wdOrigin=BROWSELINK